D.10. Accelerating Mathematical Insight: Boosting Grokking Through Strategic Data Augmentation

This idea was proposed in the 12th iteration of a Sonnet 3.5 run.

Idea

```
"Name": "data_augmentation_grokking",
"Title": "Impact of Data Augmentation on Grokking Dynamics in Mathematical
Operations",
"Experiment": "Modify AbstractDataset to include methods for operand
reversal (for addition and multiplication) and operand negation (for
addition, subtraction, and division) augmentations. Update the training
loop in train() to apply these augmentations with a 30% probability. Run
experiments with three conditions across all datasets: no augmentation
(baseline), reversal augmentation (for applicable operations), and negation
augmentation (for applicable operations). Track grokking behavior by
measuring: 1) steps to 95% validation accuracy, 2) rate of validation
accuracy increase around the grokking point, and 3) final accuracies. Plot
learning curves and gradient norm evolution for each condition. Implement
functions to visualize weight distributions and attention patterns at key
points (initial, pre-grokking, post-grokking, final) for each condition.
Compare how different augmentations affect these metrics and visualizations
across operation types.",
"Interestingness": 9,
"Feasibility": 9,
"Novelty": 8,
"novel": true
```

Link to code: https://github.com/SakanaAI/AI-Scientist/tree/main/example_papers/data_augmentation_grokking.

ACCELERATING MATHEMATICAL INSIGHT: BOOSTING GROKKING THROUGH STRATEGIC DATA AUGMENTATION

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Paper under double-blind review

ABSTRACT

This paper investigates the impact of data augmentation on grokking dynamics in mathematical operations, focusing on modular arithmetic. Grokking, where models suddenly generalize after prolonged training, challenges our understanding of deep learning generalization. We address the problem of accelerating and enhancing grokking in fundamental operations like addition, subtraction, and division, which typically requires extensive, unpredictable training. Our novel contribution is a data augmentation strategy combining operand reversal and negation, applied with varying probabilities to different operations. Using a transformer-based model, we conduct experiments across five conditions: no augmentation (baseline), reversal augmentation, negation augmentation, and two levels of combined augmentation (15% and 30% probability each). Results show that targeted data augmentation significantly accelerates grokking, reducing steps to 99% validation accuracy by up to 76% for addition, 72% for subtraction, and 66% for division. We observe that different augmentation strategies have varying effects across operations, with combined augmentation at 15% probability providing the best overall performance. Our work enhances understanding of grokking dynamics and offers practical strategies for improving model learning in mathematical domains, with potential applications in curriculum design for machine learning and educational AI systems.

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1 INTRODUCTION

Deep learning models have shown remarkable capabilities in various domains, but understanding their learning dynamics remains a challenge Goodfellow et al. (2016). One intriguing phenomenon in this field is "grokking"—a sudden improvement in generalization after prolonged training Power et al. (2022). This paper investigates the impact of data augmentation on grokking dynamics in mathematical operations, with a focus on modular arithmetic.

Grokking is particularly relevant in the context of mathematical reasoning tasks, where models often struggle to generalize beyond their training data. Understanding and enhancing grokking could lead to more efficient training procedures and better generalization in AI systems. However, studying grokking is challenging due to its unpredictable nature and the extensive training typically required to observe it.

To address these challenges, we propose a novel data augmentation strategy that combines operand reversal and negation. Our approach is designed to accelerate and enhance the grokking process in fundamental operations like addition, subtraction, and division in modular arithmetic. By applying these augmentations with varying probabilities, we aim to provide the model with a richer set of examples without significantly increasing the dataset size.

We conduct experiments using a transformer-based model Vaswani et al. (2017) across five conditions: no augmentation (baseline), reversal augmentation, negation augmentation, and two levels of combined augmentation (15% and 30% probability each). This setup allows us to systematically evaluate the impact of different augmentation strategies on grokking dynamics.

Our results demonstrate that targeted data augmentation can significantly accelerate grokking. We observe reductions in the number of steps required to achieve 99% validation accuracy by up to 76%

for addition and 72% for subtraction. Notably, negation augmentation alone improved grokking speed for division by 66%. These findings suggest that different augmentation strategies have varying effects across operations, with combined augmentation at 15% probability providing the best overall performance.

The main contributions of this paper are:

- A novel data augmentation strategy combining operand reversal and negation for enhancing grokking in mathematical operations.
- Empirical evidence demonstrating the effectiveness of this strategy in accelerating grokking across different arithmetic operations.
- Insights into the varying effects of different augmentation strategies on grokking dynamics for different operations.
- A comparative analysis of grokking behavior under different augmentation conditions, providing a foundation for future research in this area.

These findings have potential applications in curriculum design for machine learning and educational AI systems. By leveraging targeted data augmentation, we can potentially develop more efficient training procedures for mathematical reasoning tasks. Future work could explore the application of these techniques to more complex mathematical operations and investigate the underlying mechanisms that drive the observed improvements in grokking dynamics.

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2 Background

Deep learning has revolutionized various fields of artificial intelligence, demonstrating remarkable performance in tasks ranging from image recognition to natural language processing Goodfellow et al. (2016). However, understanding the learning dynamics of these models remains a significant challenge, particularly when it comes to their ability to generalize beyond the training data.

"Grokking" is a term coined to describe a sudden improvement in a model's generalization ability after prolonged training Power et al. (2022). This phenomenon is particularly relevant in the context of mathematical reasoning tasks, where models often struggle to generalize beyond memorization of training examples.

Transformer models Vaswani et al. (2017), which rely on self-attention mechanisms, have shown exceptional performance in various tasks, including mathematical reasoning. Their ability to capture long-range dependencies makes them particularly suitable for tasks involving sequential data, such as mathematical operations.

Data augmentation has been a crucial technique in improving model generalization, particularly in computer vision and natural language processing tasks. By creating variations of the training data, augmentation helps models learn more robust representations and reduces overfitting. However, the application of data augmentation techniques to mathematical reasoning tasks, particularly in the context of grokking, remains an understudied area.

Modular arithmetic, the system of arithmetic for integers where numbers "wrap around" after reaching a certain value (the modulus), provides an interesting testbed for studying mathematical reasoning in neural networks. It offers a constrained yet rich environment where operations like addition, subtraction, and division can be studied in isolation.

2.1 Problem Setting

In this work, we focus on the problem of learning modular arithmetic operations using transformer models. Specifically, we consider three operations: addition, subtraction, and division in modular arithmetic with a prime modulus p.

Let $\mathbb{Z}_p$ denote the set of integers modulo p. For any $a, b \in \mathbb{Z}_p$, we define the following operations:

- Addition: $a + b \equiv c \pmod{p}$
- Subtraction: $a - b \equiv c \pmod{p}$